# Six math clues · answer key

A Coffee Filter Parachute

Grade 2 math. Compare tens before ones; for rounding, 5 or more rounds up and 4 or less rounds down.

### Math 1 · 42

28 < 42. Two tens are fewer than four tens, so the greater total is 42 seconds.

### Math 2 · 76

76 > 67. Seven tens are more than six tens, so the greater length is 76 centimeters.

### Math 3 · 48

48 = 48. Both totals match, so the shared number is 48 drops.

### Math 4 · 20

23 has 3 ones, so round down to 20 paperclips.

### Math 5 · 60

57 has 7 ones, so round up to 60 centimeters.

### Math 6 · 100

95 has 5 ones, so round up to 100 centimeters.

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Grade 2 math. Six tasks compare two-digit values or round a two-digit value to the nearest ten.

# Reading & writing · answer key

A Coffee Filter Parachute

### Reading · 1

The flat filter has more surface area facing the air, so it makes more drag and falls more slowly than the same filter crumpled.

### Reading · 2

Several drops reduce the chance that one wobble or unusual release will decide the result.

### Reading · 3

Gravity pulls the parachute down while drag pushes against its motion and slows the descent.

### Writing

Accept a spoken, dictated, drawn, or written prediction that names one changed feature and one measurement.

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Gravity pulls a parachute down while air resistance, or drag, pushes against its motion. A wide canopy creates more drag; weight, canopy area, shape, release, and air currents can all affect a test, so change one thing at a time and repeat drops. Stay beside the child, use scissors only with grown-up help, and drop only from standing height or from a grown-up's raised hand—never climb on chairs, stair railings, windows, or balconies. Keep paperclips and small toys away from mouths.

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Offer dictation: let the child say a complete thought while you write it, then invite them to add a picture, label, or sentence.

# Four parachute clues

A Coffee Filter Parachute

### Quiz 1. What does air resistance do to a falling parachute?

Air resistance, or drag, pushes against the parachute's motion and slows its fall.

### Quiz 2. Why does a flat coffee filter usually fall more slowly than the same filter crumpled?

The wide filter shows more surface area to the air, creating more drag while its weight stays the same.

### Quiz 3. Which plan makes a fair canopy-size test?

Change only canopy size, keep weight and strings equal, and repeat the drops.

### Quiz 4. What can a tiny centered hole at the canopy top do?

A tiny top hole can let some air through more evenly, which may reduce wobble.

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